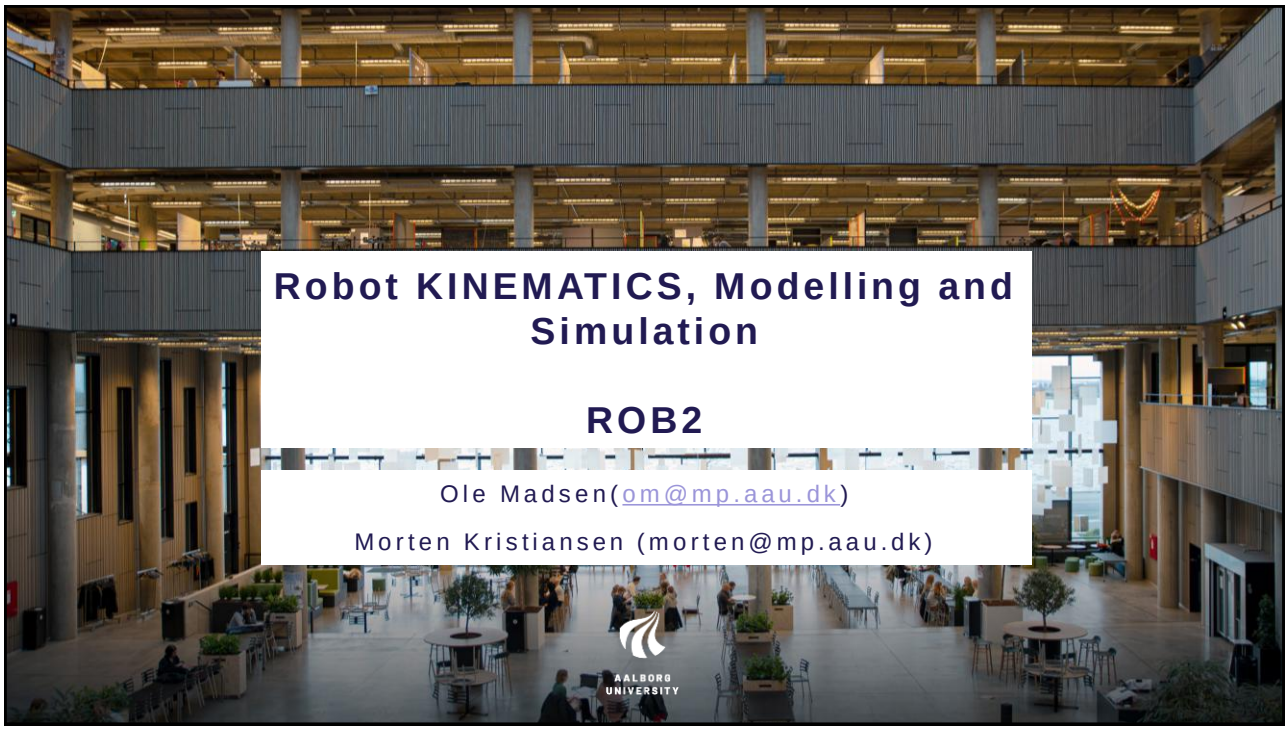

Robot KINEMATICS, Modelling and Simulation

ROB2

Ole Madsen(om@mp.aau.dk)

Morten Kristiansen (morten@mp.aau.dk)

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Fra studieordningen

- VIDEN
  - Grundlæggende aspekter i forbindelse med **robot kinematik**
  - Metoder rumlig beskrivelse af objekter
  - Grundlæggende metoder til kinematisk modellering af robotmanipulatorer
  - Principper for kinematisk robotsimulering
  - Omdannelse af bevægelser i opgaverum til robotbevægelser
- FÆRDIGHEDER
  - Anvendelse af homogene transformationsmatricer til at repræsentere position og orientering af objekter
  - Beskrive den direkte og inverse kinematik af en robot
  - Design enkle baneplanlæggere, herunder kartesiske og fælles interpolatorer
  - Programmer en industrirobot til at udføre forskellige produktionsopgaver
  - Omdanne beskrivelser i opgaverum til robotbevægelser
  - Simulere den kinematiske opførsel af en robot
- KOMPETENCER
  - Skal kunne programmere en robot, så den ønskede kinematiske adfærd opnås
  - Skal kunne simulere robottens kinematik
  - Skal kunne løse enkle produktionsopgaver med en industrirobot

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Lecture plan

No	Responsible	Contents
1	Ole Madsen	Introduction to: • the course; • robotics and robot terminology.
2	Ole Madsen	Spatial descriptions and transformation matrices
??	Ole Madsen + More	Practical exercise with the on-line programming (1.5 timer/gruppe).
4	Ole Madsen	Orientation
5	Ole Madsen	Forward Kinematics I
6	Ole Madsen	Forward Kinematics II (go though 6 DOF robot) – exercise, you go though your robot
7	Morten Kristiansen	Inverse kinematics I
8	Morten Kristiansen	Inverse kinematics II (go through 6DOF robot) – you start on your robot
9	Morten Kristiansen	Trajectory generation and control (joint)
10	Morten Kristiansen	Trajectory generation and control (cartesian)
11	Morten Kristiansen	Jacobian/Exam preparation

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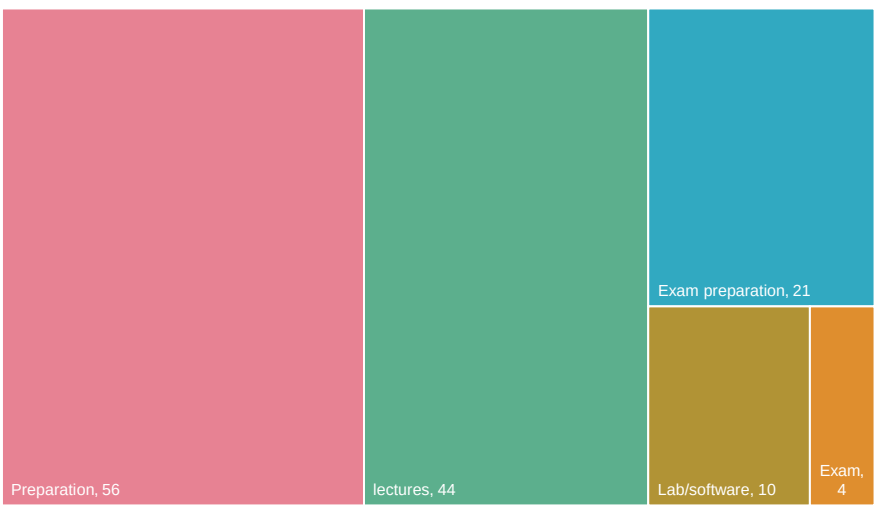
Course workload 5 ECTS (135 hours)



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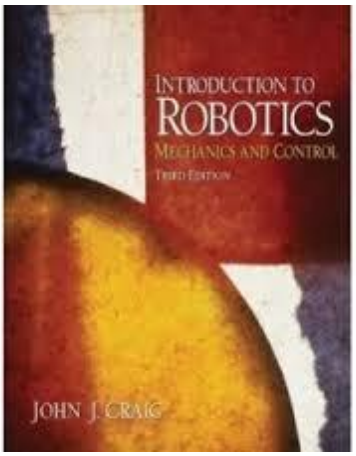
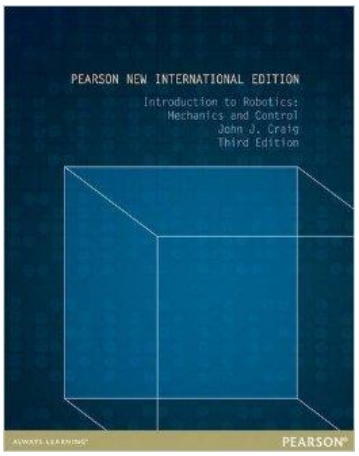
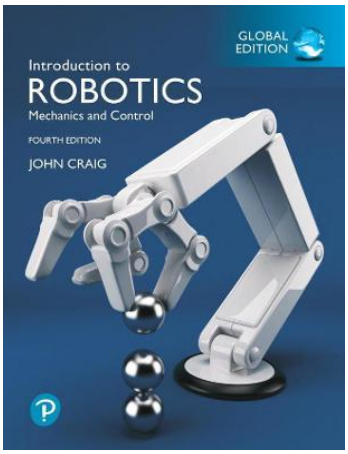
The book







BASIC ROBOTIC BOOK
(WE USE CHAPTER 1,2,3,4,5,7,12)
(OTHER CHAPTERS ON ROB3-4)

But first make sure you have installed …

- Python:
  - Examples in course runs under **Python version: 3.8.5**
  - Install IDE (e.g PyCharm)
  - Install packages (see requirements.txt under 1 lecture)
    - This will include the installation of **Peter Corkes Robotics Toolbox for Python**, information can be found: https://petercorke.github.io/robotics-toolbox-python/intro.html
    - ROBODK interface

- ROBODK:
  - Install newest version of ROBODK (https://robodk.com/download)
    - License file found on Moodle
    - **To learn: run tutorial**

Agenda

- Introduction to robotics
- Some robot terminology
- Robot components
- Robot typologies
- Selecting an industrial robot arm

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WHAT IS A ROBOT ?

9

An Industrial robot is:

- properly not the most optimal employee since it is:
  - Stupid
  - Blind
  - One-armed

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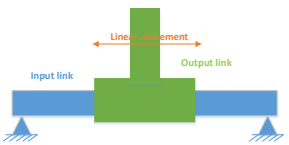
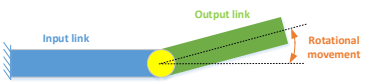
A robot is:

- An actuated **mechanism programmable** in two or more **axes** with a degree of **autonomy moving** within its environment, to perform intended tasks.

- Axis:
  - Direction used to specify the robot motion in a **linear** or **rotary** mode.





DS/ISO 8373 (1. EDITION 2013-06-04)
MANIPULATING INDUSTRIAL ROBOTS - VOCABULARY

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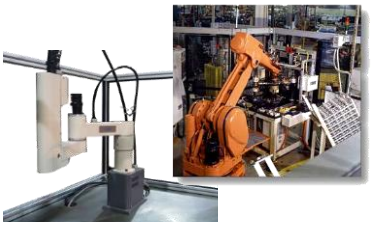
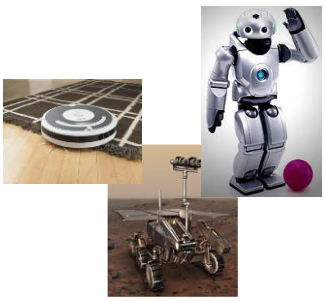
Industrial robots vs service robots

- Industrial robots
  - Work in relatively stable and structured environments
  - Limited mobility
  - Relative simple control programs
  - Work inside fences



- Service robots
  - Work in the “real” world
  - Are mobile
  - Require a high degree of autonomy and intelligence
  - Co-exists and collaborate with humans



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Why automate ?

- Reduce costs
- Improve working environment:
  - Remove dangerous jobs
  - Eliminate repetitive jobs
- Improve product quality
- Reduce through-put time
- Carry out processes and features that cannot be made manual

Why use robots for automation ?

- Robot based solutions are more flexible than traditional automation:
  - Kinematic flexibility
  - Fast change over to new products
  - Flexible capacity

Why not automate ?

- High initial cost.
- Time to implement.

- The product life cycle is short.
- The products are highly customized.
- There are large variation in demands.

- The task is too technical difficult to automate. E.g:.
  - Problems with physical access to the work location,
  - Part location is unstructured
  - Adjustments required in the tasks
  - Manual dexterity requirements
  - Demands on hand eye coordination.

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Challenges for you $$$$$$$

- Reducing installation costs
- Improving flexibility – it takes time to reinstruct a robot for a new task.
- Improving capabilities – there are many tasks that robots cannot efficiently solve
- MAKE-IT-EASY – it still requires special skills to program a robot
- Improving abilities to adapt to changes in the environment
- Find a way to fire a robot if there is no use for it !!

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Agenda

- Introduction to robotics
- **Some robot terminology**
- Robot components
- Robot typologies
- Selecting an industrial robot

Joints axes and links

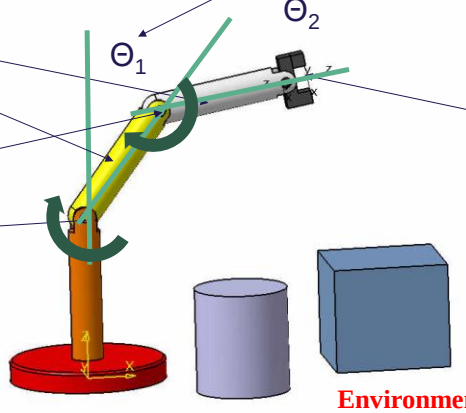
Joint values

Link:

Rigid body connecting neighboring joints

Joint axes:

Directions used to specify the motion in a linear or rotary mode



End-effector = tool

Environment

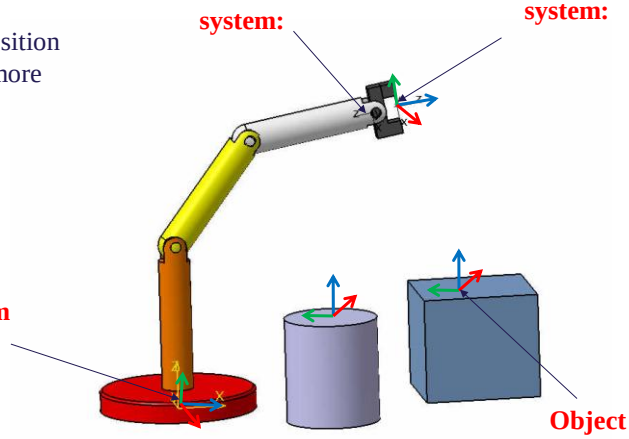
Coordinate systems

Coordinate systems:

Are used to represent the position and orientation of objects (more later)

Wrist coordinate system:

Tool coordinate system:



Base coordinate system

Object (or user) coordinate system:

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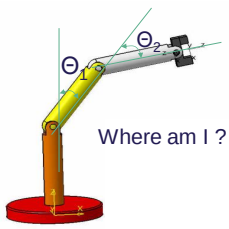
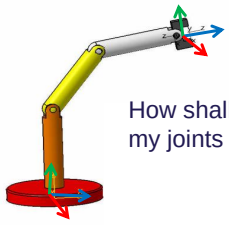
Forward and inverse kinematics

- Forward kinematics:
  - Θ_1 and Θ_2 are known
  - Compute the location of the tool



Where am I ?

- Inverse kinematics:
  - The location of the tool is known.
  - Compute Θ_1 and Θ_2



How shall I put my joints ?

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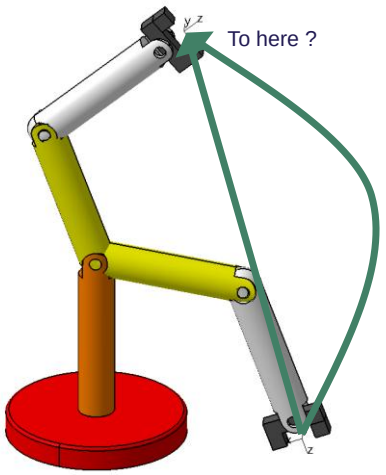
Trajectories



Program

```
MODULE SVEJS
PROC main()
   MoveJ,*,v100,z50,tool0;
   ArcL\On,*,v200,sm1,wd1,wv1,fine,tool0;
   ArcL\Off,*,v100,sm1,wd1,wv1,fine,tool0;
   MoveL,*,v200,fine,tool0;
  ENDPROC
ENDMODULE
```

Controller

Arm

Control Data

Agenda

- Introduction
- Some robot terminology
- **Robot components**
- Robot typologies
- Robot programs and programming

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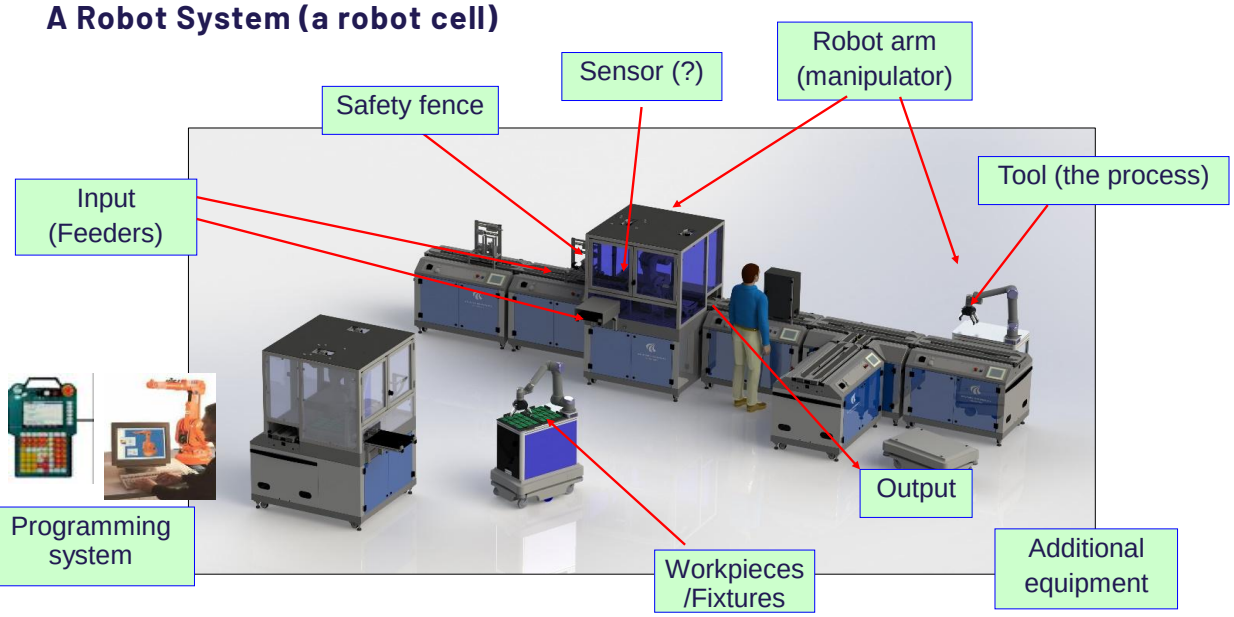
A Robot System (a robot cell)



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Agenda

- Introduction to robotics
- Some robot terminology
- Robot components
- **Robot typologies**
- Selecting an industrial robot

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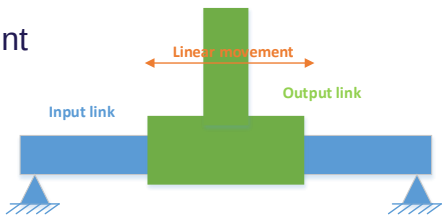
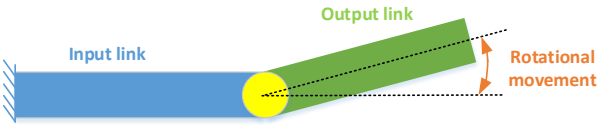
Joint types

Translational joint



Rotational joint



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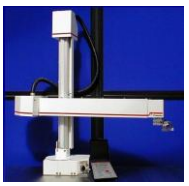
Kinematic configurationa

RRRRRR

TTT



RTT

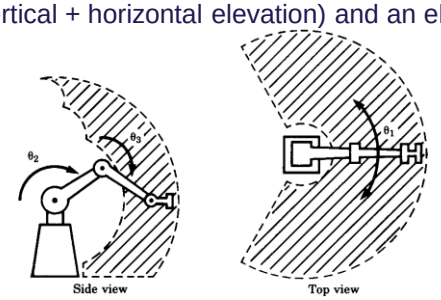
RRT

RRR

SCARA:RRTR

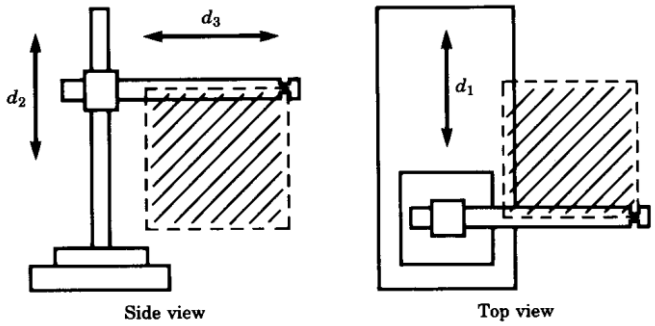
Articulated (anthropomorphic) (RRR)

- Two shoulder joints (vertical + horizontal elevation) and an elbow joint parallel to the elevation joint.



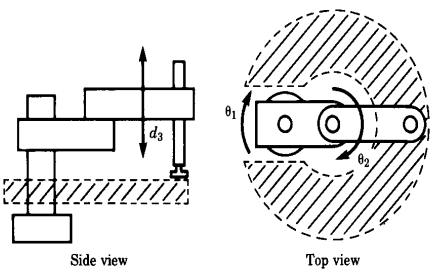
Cartesian (gantry) robot (TTT)

- Three perpendicular translational axes.
- Stiff structures allow



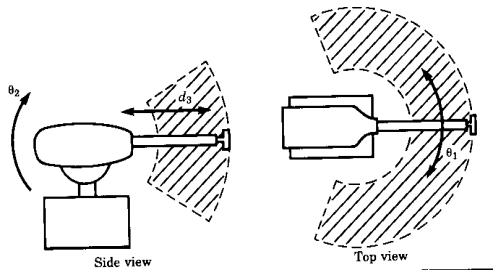
Scara (RRTR)

- Three revolute parallel joints allowing it to move and orient in plane.
- Usually very fast robots.
- Well suited to pick and place.

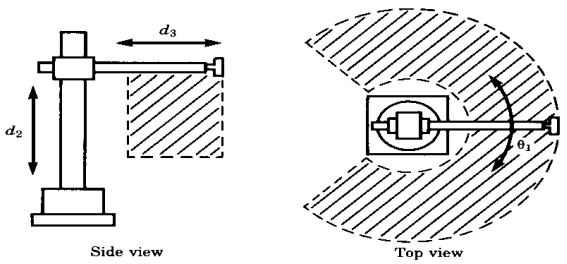


More configurations

Spherical (RRT)

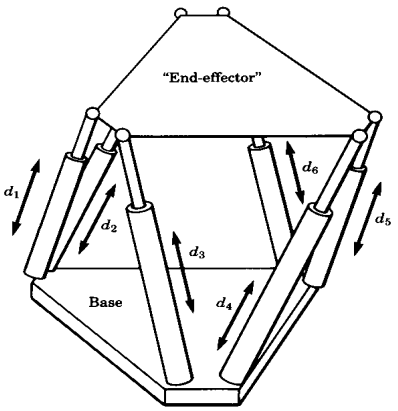


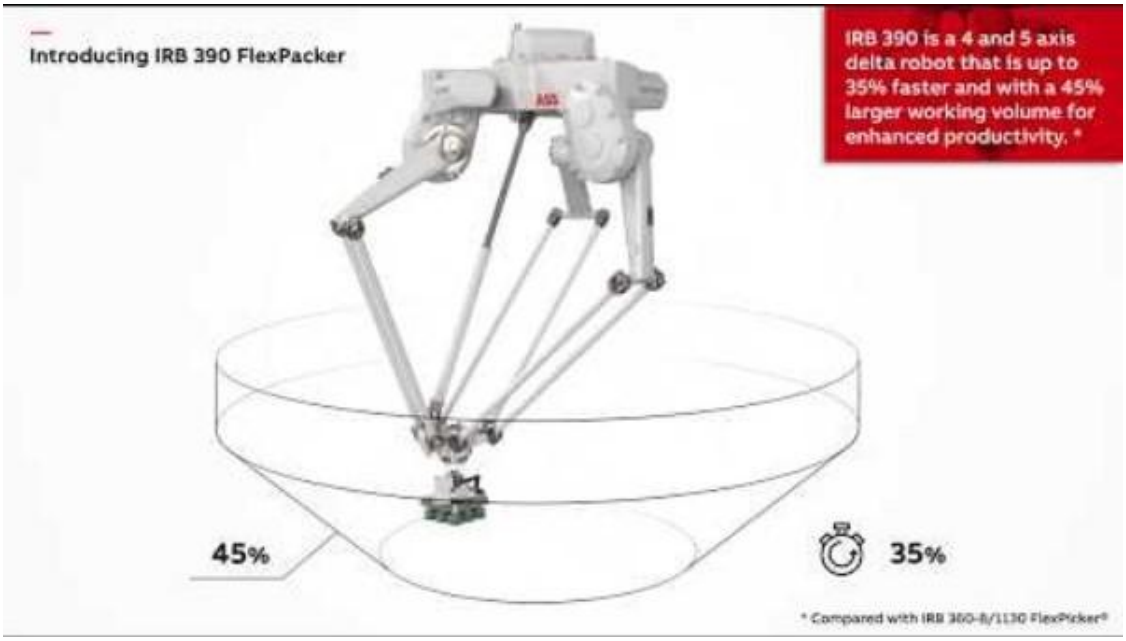
Cylindrical (RTT)



Closed structures

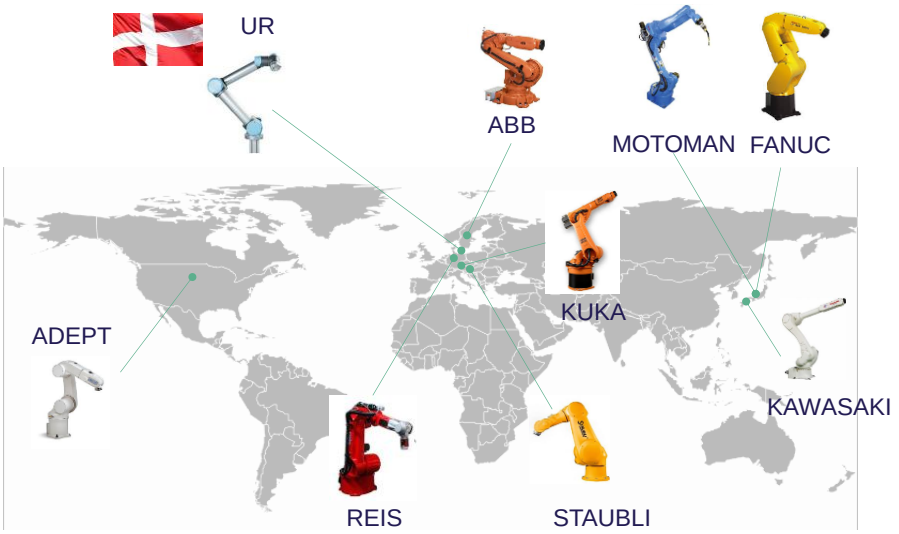
- Increased stiffness
- Fast and/or strong robots
- Reduced allowable range of motion





Agenda

- Introduction to robotics
- Some robot terminology
- Robot components
- Robot typologies
- **Selecting an industrial robot arm**



KUKA-robots

KUKA

KR 3 R540

Technical data

Maximum reach	541 mm
Maximum payload	3 kg
Pose repeatability (ISO 9283)	± 0.02 mm
Number of axes	6
Mounting position	Floor; Ceiling; Wall
Footprint	179 mm x 179 mm
Weight	approx. 26.5 kg

Axis data

Motion range	
A1	±170 °
A2	-170 ° / 50 °
A3	-110 ° / 155 °
A4	±175 °
A5	±120 °
A6	±350 °

Operating conditions

Ambient temperature during operation	5 °C to 45 °C (278 K to 318 K)

Protection rating

Protection rating (IEC 60529)	IP40
Protection rating, in-line wrist (IEC 60529)	IP40

Controller

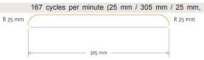
Controller	KR C4 compact

Teach pendant

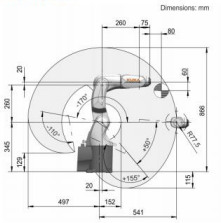
Teach pendant	KUKA smartPAD

Cycle time

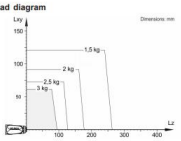
167 cycles per minute (25 mm / 305 mm / 25 mm, 1 kg)



Workspace graphic

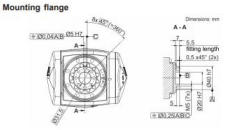


Payload diagram



The KR 3 R540 is designed for a rated payload of 2 kg in order to optimize the dynamic performance of the robot. With reduced load center distances, higher loads up to the maximum payload may be used. The specific load case must be verified using KUKA.Load. For further consultation, please contact KUKA Support.

Mounting flange



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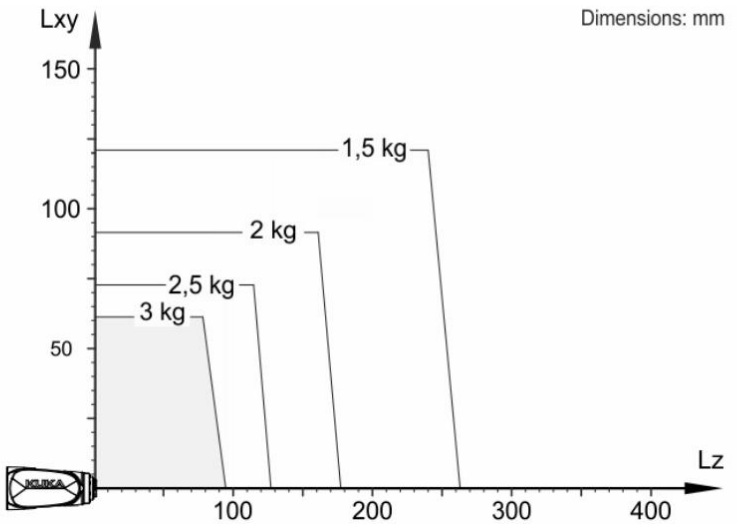
Payload diagram



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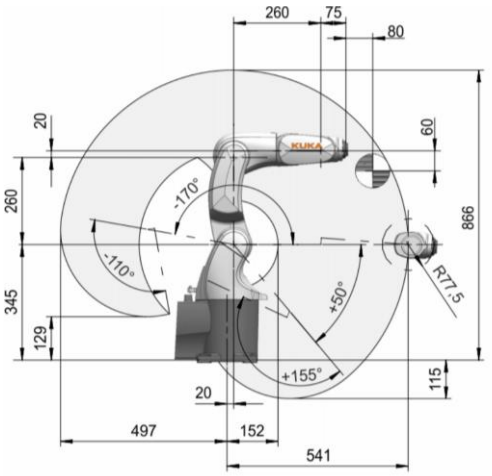
Working envelope/reach



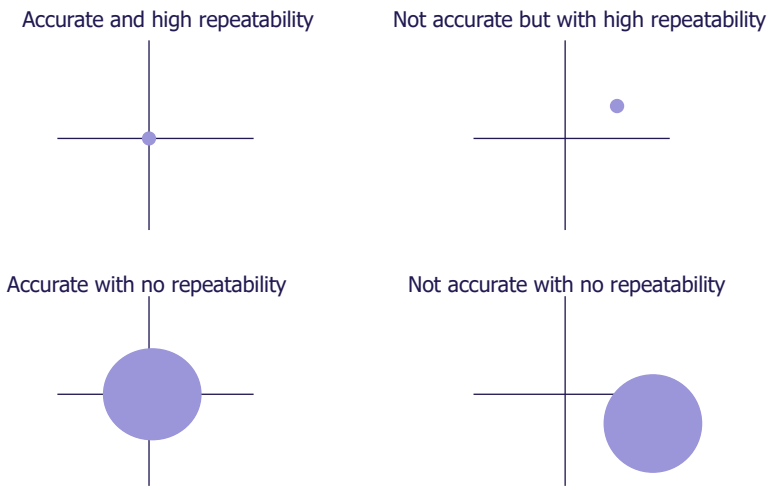
FOR KUKA HOME PAGE

Robot Accuracy and Repeatability

Two terms used to define precision in robotics:

1. Accuracy - capability to position the robot's wrist at a desired location in the work space
2. Repeatability - capability to position the wrist at a previously taught location in the work space

Robot Accuracy and Repeatability



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Speed/Cycle time

- Max. speed
  - Axis 1 150°/s
  - Axis 2 150°/s
  - Axis 3 150°/s
  - Axis 4 360°/s
  - Axis 5 360°/s
  - Axis 6 450°/s
- Cycle time: 138 cycles/min (25/305/25; 1 kg Payload)

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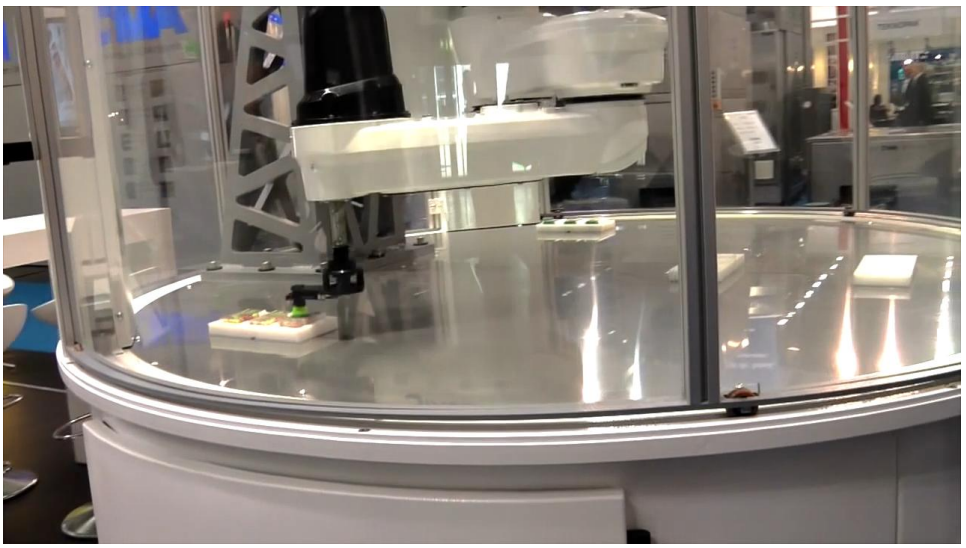

IP rating

- IP ratings designate the degree of protection an enclosure provides against the ingress or intrusion of foreign objects.
- IP ratings consist of the letters “IP” followed by two numbers:
  - The first number designates protection against solid objects.
  - The second number designates protection against water.

IP Rating	Solid Object Size
IP1X	≥ 50mm (1.97”) diameter
IP2X	≥ 12.5mm (0.49”) diameter
IP3X	≥ 2.5mm (0.098”) diameter
IP4X	≥ 1.0mm (0.039”) diameter
IP5X	dust-protected
IP6X	dust-tight

IP Rating	Type of Water
IPX1	vertical dripping
IPX2	dripping (up to 15º tilt)
IPX3	spraying (up to 60º angle)
IPX4	splashing from any direction
IPX5	jets from any direction
IPX6	powerful jets from any direction
IPX7	temporary immersion
IPX8	continuous immersion

Collaborative or traditional ??

ISO 10218 (Parts I and II)

ISO/TS 15066:2016

Force limited

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KUKA

KR 3 R540

Technical data

Maximum reach	541 mm
Maximum payload	3 kg
Pose repeatability (ISO 9283)	± 0.02 mm
Number of axes	6
Mounting position	Floor; Ceiling; Wall
Footprint	179 mm x 179 mm
Weight	approx. 26.5 kg

Axis data

Motion range	
A1	±170 °
A2	-170 ° / 50 °
A3	-110 ° / 155 °
A4	±175 °
A5	±120 °
A6	±350 °

Operating conditions

Ambient temperature during operation	5 °C to 45 °C (278 K to 318 K)

Protection rating

Protection rating (IEC 60529)	IP40
Protection rating, in-line wrist (IEC 60529)	IP40

Controller

Controller	KR C4 compact

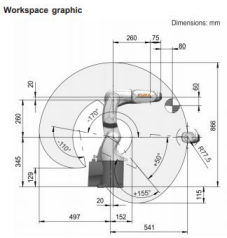
Teach pendant

Teach pendant	KUKA smartPAD

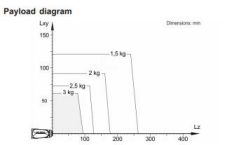
Cycle time

167 cycles per minute (25 mm / 305 mm / 25 mm, 1 kg)

Workspace graphic

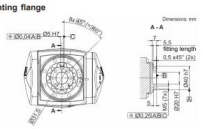


Payload diagram



The KR 3 R540 is designed for a rated payload of 2 kg in order to optimize the dynamic performance of the robot. With reduced load center distances, higher loads up to the maximum payload may be used. The specific load case must be verified using KUKA.Load. For further consultation, please contact KUKA Support.

Mounting flange



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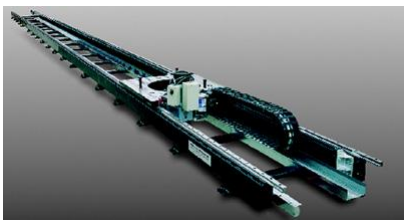
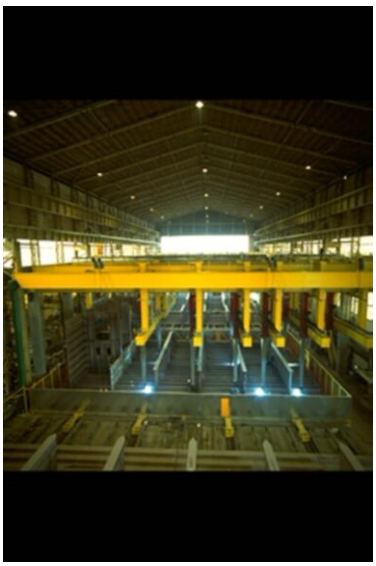
External axes





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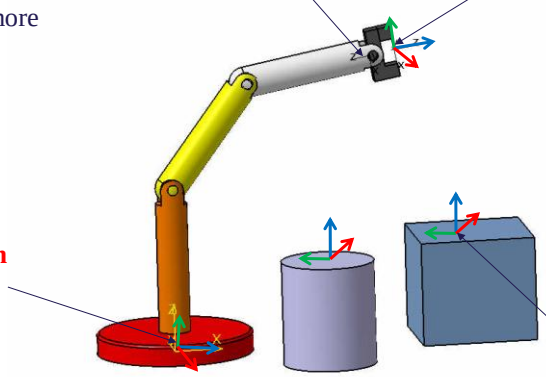
Next time

Coordinate systems:

Are used to represent the position and orientation of objects (more later)



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Lecture plan

No	Responsible	Contents
1	Ole Madsen	Introduction to: • the course; • robotics and robot terminology.
2	Ole Madsen	Spatial descriptions and transformation matrices
??	Ole Madsen + More	Practical exercise with the on-line programming (1.5 timer/gruppe).
4	Ole Madsen	Orientation
5	Ole Madsen	Forward Kinematics I
6	Ole Madsen	Forward Kinematics II (go though 6 DOF robot) – exercise, you go though your robot
7	Morten Kristiansen	Inverse kinematics I
8	Morten Kristiansen	Inverse kinematics II (go through 6DOF robot) – you start on your robot
9	Morten Kristiansen	Trajectory generation and control (joint)
10	Morten Kristiansen	Trajectory generation and control (cartesian)
11	Morten Kristiansen	Jacobian/Exam preparation

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Exercise:

1. ROBODK
   a) Install RoboDK
   b) Run the tutorial
2. Python
   a) Make sure you have installed Python (between 3.8 and 3.10)
   b) Install: requirements.txt
   c) Run test program
3. Discuss:
   a) How do we represent robot locations ?
   b) What is forward kinematics ?
   c) What is inverse kinematics ?
   d) What is a robot trajectory ?
4. Investigate various robots:
   a) Find data sheets for:
      - KR6 700sixx (KUKA)
      - UR 5 (Universal robot)
      - Adept Cobra 800 (Omron/Adept)
   b) Identify the joint axes and the links of the robots
   c) Identify the robot topology
   d) Compare the robots (workspace, payload, speed, weight..)

Hand-in solutions for 1.4 (Send by mail to me)

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